

Casinc Project architecture

Casinc current state relies all the execution off-chain with only deposits, withdrawals on chain. This can be further improved with following requirements

Casinc contract

This is the main contract that will provide the game for the users, deposit betting amount, place bets and allow admins to control the game state and withdrawal requests from the user.

Functionality

Create Deposit

Before user can participate in the game, they would first need to deposit amount from User to Casinc contract. The deposit is transferred from User account to Casinc contract account. User can deposit more if needed.

Play Casinc

- Once the user contract is created, they can use the deposited amount to make a bet. This function also take betting parameters like multiplier etc..
- Contract places the bet and run the game.
- If the user won the bet, the winning amount is stored in the Winnings account. These winnings are accessed by the User after some time. This provides admin necessary time to validate the User winnings

Unlock Winnings

- Once the set time is elapsed, users can choose to initiate a withdrawal of the winnings or move the winnings to deposit to place more bets.
- If user choose to withdraw winnings, then a new Withdrawal request is created for the Admins to Accept the withdrawal.

Execute withdrawal

- Once the admin accept the withdrawal request, user can request the winnings from Casinc contract to their account.

Admin contract

Admin contract enable certain admin functionality on the Casinc. Admin contract is a multisig contract and all the requests going through the Admin contract will need a multisig approval. This multisig can be either external multisig like Gnosis Safe or regular Account multisig.

Set Game parameters

Admins can set the game parameters at any time. The new game parameters can be executed immediately or executed after a specific block.

Decide Withdrawl request

- When a user initiate a withdrawal request, Admins has the ability to verify their winnings and either approve or reject Withdrawal request
- Before approving the contract ensures Casinc contract has enough balance that can transferred to user when executed.

The project will overall contain two contracts

- Casinc contract
- Admin Contract

Casinc contract knows the special privilege of Admin contract to do necessary actions such as Deciding the withdrawal requests, Setting game parameters, blocking specific user from playing.

Casinc contract is the entry point for the users to play the game by betting their deposits for higher winnings.

Architecture Diagram

